Technology Stack

Date	16 September 2026
Team ID	SWTID-2026-2630
Project Name	AI Fit Track API
Maximum Marks	2 Marks

Technology Stack Details

AI Fit Track is a headless RESTful backend; the "client" during development and grading is an API tool (Thunder Client / Postman) rather than a browser UI, with a `client/` folder reserved for a future React or mobile frontend.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Thunder Client (VS Code extension) & Postman — used as the request/response client; the project's `client/` folder is scaffolded for a future React/mobile UI but out of scope for this cycle	Project scope for 15–29 Sept 2026 is a backend/API service; API clients are sufficient to exercise and grade every endpoint via the shared collection
2	Backend / Server-Side	Node.js + Express.js, with `cors` enabled	Non-blocking I/O suits the Gemini AI calls; minimal boilerplate for fast REST route development; whole team already proficient in JavaScript
3	Database / Data Storage	MongoDB with Mongoose ODM	Schema-flexible NoSQL store fits the User/Workout documents; Mongoose adds field validation (required, unique, minlength, enum, min), defaults and `timestamps` on top of raw MongoDB
4	Cloud / Hosting / Deployment	API server run locally on port 5000 (`process.env.PORT || 5000`) with MongoDB at `mongodb://localhost:27017/aifittrack` for the grading window; fully `.env`-driven configuration so the API itself is cloud-deploy-ready	Keeps the grading environment simple and reproducible while leaving a clear path to a hosted MongoDB service and a cloud deploy without code changes
5	Version Control & CI/CD	Git & GitHub – github.com/thamaraiselvan6/fittrack_team_projt	Single source of truth for the 5-member team; commit history traces each Epic/Sprint
6	Third-Party APIs / Other Tools	Google Gemini AI (workout recommendation & fitness insights via `geminiService.js`), `jsonwebtoken` + `bcryptjs` (auth/security), `dotenv` , `cors` , `nodemon` (dev)	Each library enables one specific, named feature of the solution — generative insight, secure auth, config isolation, and cross-origin support